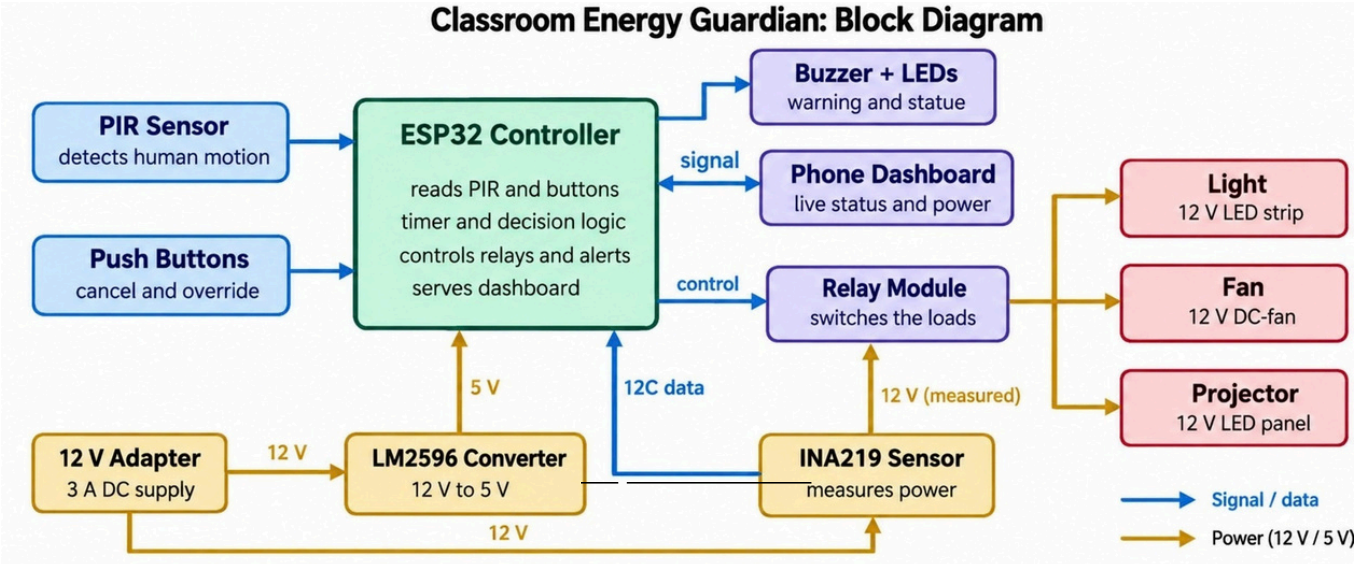
Classroom Energy guardian-Detect empty classrooms and automatically manages light/fan

Methodology

Our prototype uses hardware-based automation to manage classroom energy without coding or Wi-Fi. The system operates through three sequential phases: Detection, Signal Regulation, and Power Switching.

1. *Detection Phase:* The PIR sensor detects human presence by sensing infrared changes and sends a HIGH signal to the ESP32 whenever motion is detected.
2. *Signal Regulation Phase* : When motion is detected, the sensor outputs a 3.3V HIGH signal to a hardware-based Delay Relay Module. This hardware timer regulates the signal duration using an onboard potentiometer (variable resistor). It ensures the power stays on during brief periods of stillness, preventing erratic switching.
3. *Power Switching Phase*

The timed signal acts as an electronic trigger, instantly energizing the relay's internal electromagnet. This magnetic pull snaps the switch shut, completing the electrical circuit and flooding the classroom appliances with power.
Immediately, the LED lights illuminate the room and the fans spin to life, creating an instantly optimized learning environment. The moment the room is completely empty and the hardware timer runs out, the circuit safely snaps back open—cutting all power instantly to eliminate energy waste and guard the school's electricity.



Proposed outcomes

• *Immediate Energy Savings* : Dramatically cuts electricity consumption by ensuring lights and fans operate only when human presence is active, wiping out empty-classroom waste.
• *Hardware-Level Security:* Achieves zero vulnerability to cyber threats. Because it operates completely offline without Wi-Fi or coding , the system is 100% immune to hacking or network downtimes.

• *Highly Cost-Effective & Scalable:* Offers an ultra-low-cost setup using inexpensive, off-the-shelf hardware components, making it affordable for schools to deploy across hundreds of classrooms.
• *Zero-Maintenance Operation* : Eliminates the need for software updates, it creating a true "plug-and-play" system.